



A Neural Codec Approach for Noise-Robust Bandwidth Expansion

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Abstract

Neural audio codec technology has recently attracted significant attention in various speech processing tasks due to its efficient quantized latent features. In this work, we introduce a novel approach that leverages a pre-trained neural codec network to perform both speech denoising and bandwidth expansion simultaneously. Specifically, we design a conformer-based deep neural network to predict clean codebook indices, which are then used by the pre-trained audio codec model to generate enhanced and bandwidth-expanded audio. We investigated several strategies for generating the clean indices and compared our approach with state-of-the-art methods on the Valentini-Botinhao noisy test set. Experimental results demonstrate that our method achieves performance comparable to leading approaches in noise-robust bandwidth expansion tasks while offering promising improvements in the quality and intelligibility of narrow-band signals. Audio samples are available.¹

Index Terms: speech enhancement, bandwidth expansion, neural codec

1. Introduction

Audio bandwidth expansion (BWE), also known as audio super-resolution, is a technique designed to restore or generate the high-frequency components of an audio signal that have been lost or were never recorded, typically in narrow-band recordings (such as traditional telephone speech). The primary goal is to enhance the perceived quality and naturalness of the voice by extending its effective frequency range. Narrow-band audio generally covers frequencies up to about 3.4 kHz, and the absence of higher-frequency information can make the voice sound muffled and less intelligible. In addition, many historical recordings suffer from background noise that further degrades speech quality. In this work, we propose a noise-robust bandwidth expansion approach that not only recovers the missing high-frequency content but also effectively suppresses background noise.

Early BWE methods employed signal processing algorithms such as linear predictive coding (LPC)[1] or rule-based spectral mapping[2] to restore the high frequency details. However, these approaches rely on linear or heuristic models that are not well-equipped to capture the complex, nonlinear relationships present in natural speech, often resulting in suboptimal reconstruction of the high-frequency components. Recent approaches leverage deep neural networks (DNNs) to achieve remarkable performance due to their powerful nonlinear modeling capabilities.

DNN-based bandwidth expansion (BWE) approaches can be broadly classified into three categories: methods operating in the Mel-spectrum domain, those working in the linear spectral domain, and those applied directly in the waveform domain. Mel-spectrum domain methods [3, 4, 5] aim to reconstruct the Mel-scale spectrum of wide-band or full-band speech and then convert the enhanced Mel-spectrum into an audio waveform using a neural vocoder [6]. These methods are computationally efficient because of using fewer frequency features compared to processing all frequency bins. However, the Mel-scale representation has lower frequency resolution in the high-frequency range, which inevitably leads to a loss of fine details. Furthermore, since the Mel-spectrum only captures amplitude information without phase components, these approaches are less effective in low signal-to-noise ratio (SNR) conditions, often resulting in speech distortion and artifacts in the high-frequency bands. To address these issues, SWiBE [7] and AERO [8] utilize the complex spectrum as the input feature to better reconstruct high-frequency components. Specifically, AERO incorporates Generative Adversarial Networks (GANs) to generate higher-quality speech, while SWiBE employs a stochastic Denoising Diffusion Process (DDP) to achieve notable improvements. On the other hand, [9] proposed a waveform-domain approach for BWE using a DDP model, and Lin et al. [10] designed a two-stage network that jointly processes narrow-band signals in both the frequency and time domains.

In contrast to methods that use spectral or waveform features, Genhancer [11] leverages quantized latent representations produced by a Neural Audio Codec (NAC) for generative speech enhancement. Specifically, it predicts clean codebook indices from noisy ones using MaskGIT and a language model (LM). NAC [12, 13, 14] compresses audio into discrete tokens through residual vector quantization (RVQ) and an encoder-decoder architecture, achieving a high compression rate without losing critical audio information. This technique has also been successfully applied to language model-based speech synthesis tasks such as Text-To-Speech (TTS) [15, 16].

Inspired by previous work, we propose a novel approach for noise-robust bandwidth expansion. Our method employs a separate model that uses continuous embeddings as input features to predict clean discrete tokens (i.e. clean codebook indices). These estimated tokens are then used by a pre-trained Neural Audio Codec (NAC) to generate the clean audio waveform. The main contributions of this paper are as follows: 1) We present a NAC based method on noise-robust bandwidth expansion. 2) We conduct a comprehensive study on the performance of clean token prediction using different types of embeddings and training strategies.

¹Audio samples: <https://winkee520.github.io/NRBWE/>

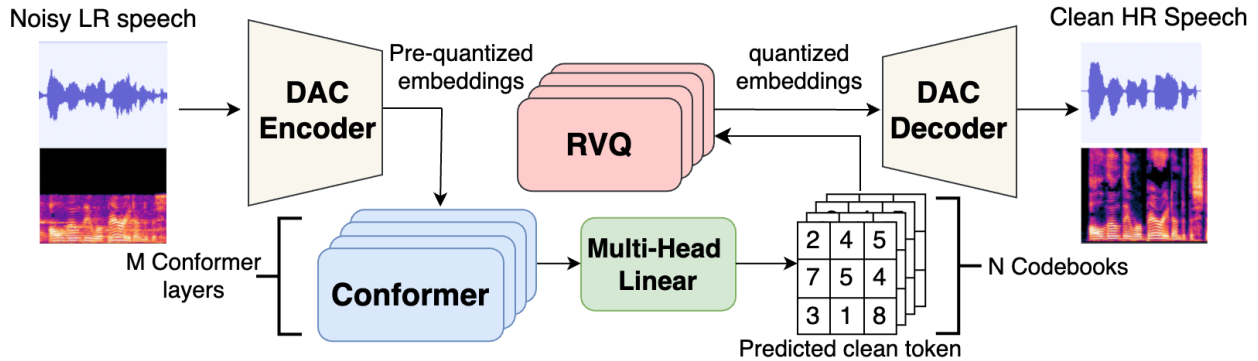


Figure 1: System overview of the proposed system. The system contains 2 parts: Descript Audio Codec (DAC) and clean token predictor. DAC is consisted of DAC Encoder, Residual Vector quantization (RVQ) and DAC Decoder, and clean token predictor has multiple Conformer layers and Multi-head linear layers.

2. Method

2.1. Method Overview

The system overview is presented in Figure 1. In this work, we focus on recovering a clean, high-resolution (HR) speech signal (sampled at 16 kHz) from a low-resolution (LR) speech signal (sampled at 8 kHz) that is degraded by background noise. To achieve this, we employ a pre-trained Descript Audio Codec (DAC)[14] as our audio waveform generator, which operates on 16 kHz audio. We propose a simple, Conformer-based clean token predictor [17] to generate predicted clean tokens (i.e., clean codebook indices). These predicted tokens are then used to produce quantized embeddings that are fed into the DAC Decoder to synthesize the reconstructed HR audio.

2.1.1. Descript Audio Codec

DAC is a high-fidelity audio compression model that converts audio waveforms to low-bitrate discrete tokens while preserving critical speech quality. Audio codecs are typically used for efficient signal transmission, but recently researchers discovered that the discrete token is more efficient than continuous latent representations on various downstream tasks such as language modeling and speech synthesis. Like other neural audio codec approaches, DAC comprises three major components: an Encoder module, a Residual Vector Quantization (RVQ) module, and a Decoder module.

The Encoder consists of a series of 1D convolutional layers that downscale the waveform to a lower rate, while the Decoder is trained to reconstruct the original audio from quantized embeddings. The core of DAC is the RVQ module, which is made up of N vector quantization processing block corresponding to N different codebooks. Similar to VQ-VAE [18], RVQ is a multi-stage process that iteratively quantizes the original signal and its residual error, with each stage refining the approximation. By breaking the quantization task into smaller, successive approximations, RVQ strikes a balance between compression rate and signal fidelity.

Because the convolutional encoder-decoder model directly processes waveforms, it may struggle to capture fine frequency details, particularly in the high-frequency range. To address this, DAC employs multi-scale STFT discriminators and a Mel-scale loss to alleviate aliasing artifacts and frequency-domain distortions, potentially making it well-suited for bandwidth expansion tasks. In this work, we use the discrete tokens from N

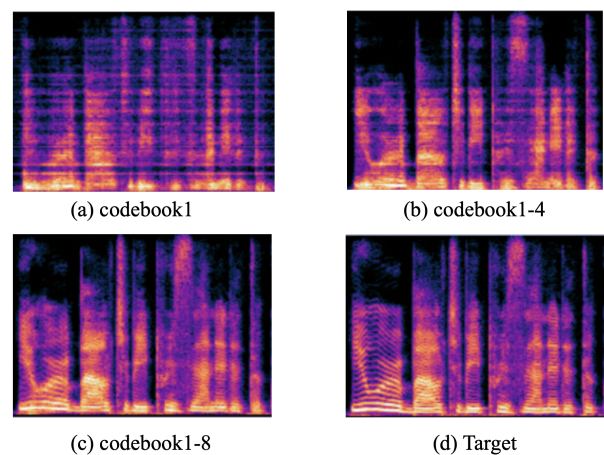


Figure 2: Spectrogram of using different number of codebooks. (a) only uses the first codebook of RVQ for decoding; (b) present result of using 3 more residual codebooks; (c) shows result of using 8 codebooks; (d) is the target

different codebooks as the training targets, enabling the clean token predictor module to generate approximate clean tokens from low-pass filtered noisy input.

2.1.2. Parallel clean token prediction

Considering the hierarchical structure of RVQ, the model quantizes and generates audio in a coarse-to-fine manner, as illustrated in Figure 2. As the number of residual codebooks increases, the reconstructed speech—encompassing content, speaker identity, and harmonic structure—becomes progressively more recognizable and intelligible. Consequently, some speech synthesis methods based on speech language models (SLMs) [15, 16] adopt an auto-regressive (AR) approach to predict target tokens, since each subsequent residual token strongly depends on the current one. In contrast, our approach predicts the clean token in a parallel manner without relying on any SLM or textual information. We believe that, given sufficient prior knowledge, the model can learn the hierarchical structure effectively. Our clean token predictor consists of two components: A multi-layer Conformer block comprising M Conformer layers

and A multi-head linear block with N heads (equal to the number of codebooks), with a ReLU activation function between each linear layer. The model is trained using cross-entropy loss to predict the target token probabilities, which is defined as:

$$\mathcal{L}_{\text{cross-entropy}} = -\frac{1}{S} \sum_{i=1}^S \sum_{c=1}^C y_{i,c} \log \hat{y}_{i,c}, \quad (1)$$

Here, S denotes the number of samples, C represents the number of classes (which, in this context, equals the codebook size), $y_{i,c}$ is the target codebook index and $\hat{y}_{i,c}$ is the predicted probability that sample i belongs to class c . The final predicted clean token is obtained by applying the argmax function along the class dimension.

To identify the optimal input features for our predictor, we experimented with both pre-quantized embeddings and the quantized embeddings generated by DAC at various stages (see Figure 1). In addition to using a standard parallel prediction approach, we implemented a progressive training strategy. In this progressive approach, we solely train the model to predict first codebook index for 10 epochs, then train to predict codebooks indices through 1 to 4 for an additional 10 epochs, and gradually cover all codebooks. This training approach aligns with the coarse-to-fine structure of RVQ, leveraging the insight that the first codebook plays a crucial role in shaping overall performance. Detailed training setup and comparative results are discussed in Sections 3 and 4.

3. Experiments

3.1. Dataset

3.1.1. DNS-challenge wide-band dataset

The DNS-challenge[19] dataset has been widely used for various DNN-based speech enhancement tasks and noise-robust speech technology. The clean speech training set is about 760 hours and contains over 10k speakers and 11 different languages. The noise dataset is collected from several resources and has total 181 hours and 150 classes. In this study, we only use DNS-challenge wide-band dataset for training, it contains about 160 hours high quality read speech in English and 50 hours noise clips.

3.1.2. Valentini-Botinhao dataset

The Valentini-Botinhao dataset[20] is another popular dataset for speech processing tasks under noise conditions. This English corpus comprising 28 speakers for training and 2 speakers for testing. The training set features 40 distinct noisy conditions, while the test set includes recordings under 20 additional noisy conditions. Since the original audio is sampled at 48 kHz, we down-sampled it to 16 kHz for this task. In our experiments, we use only the noisy test set to evaluate speech quality improvements.

3.2. Experimental setup

We use a pre-trained DAC network² trained on clean speech and music data at a 16 kHz sampling rate, which compresses a one-second audio clip into 50 frames. The total number of codebooks N used in RVQ is 12. Our clean token prediction model incorporates a Conformer block implemented in PyTorch, consisting of 8 layers with 8 attention heads. The depthwise convolution within the Conformer block uses a kernel size of 15, and

²<https://github.com/descriptinc/descript-audio-codec>

the feedforward network is configured with 1024 hidden dimensions. The model features a multi-head linear block composed of two linear layers separated by a ReLU nonlinear activation function and followed by a LayerNorm layer. The final linear layer projects the 1024-dimensional hidden state into a 12*1024 output, yielding 12 separate predictions with 1024 classes each. During training, a dropout rate of 0.1 is applied, and the model is optimized using the AdamW optimizer with an initial learning rate of 0.0003. All layer weights are initialized using the Xavier initializer.

To achieve noise-robust bandwidth expansion, we adopt a two-stage training strategy. In the first stage, the predictor is pre-trained solely on the noise suppression task. In the second stage, it is fine-tuned on a joint task that combines noise suppression with bandwidth expansion. Although the training target remains the same across both stages, the input data differs: during stage one, the input is only corrupted by background noise; during stage two, the noisy input is downsampled to 8 kHz and then upsampled back to 16 kHz using sinc interpolation. Additionally, paired training data is synthesized on the fly during each training iteration. In every training loop, we generate 24,000 clips with 2 seconds duration, and the signal-to-noise ratio (SNR) of the synthesized noisy data is randomly sampled from -5 dB to 20 dB. The progressive training method described in Section 2.1.2 is structured as follows: Epochs 1–10: Train the model to predict codebook 1; Epochs 11–20: Extend training to predict codebooks 1–4; Epochs 21–30: Extend prediction to codebooks 1–6; Epochs 31–40: Further train on codebooks 1–8; Remaining epochs: Train the model to predict all codebooks (1–12).

3.3. Evaluation Metrics

To evaluate the speech quality improvement, we use 2 popularly used metrics for this task: DNSMOS[21] and Log-Spectrum Distance(LSD). DNSMOS is a non-intrusive DNN-based speech quality assessment designed for speech enhancement task, particularly for denoising task. It has 3 scores to assess the speech quality: CSIG evaluates How natural and clear the speech sounds, CBAK can tell How intrusive the background noise is, and COVL is a combination of CSIG and CBAK. Log-Spectral Distance (LSD) is a also widely used objective metric especially for bandwidth expansion. It measures the similarity between predicted and target speech signals in the log-spectral domain. The formula of LSD is defined as:

$$LSD = \sqrt{\frac{1}{K} \sum_{k=1}^K [\log S_{\text{enh}}(k) - \log S_{\text{clean}}(k)]^2} \quad (2)$$

Where, K is the number of frequency bins; $S_{\text{enh}}(k)$ is the power spectral magnitude of the enhanced signal at frequency bin k and $S_{\text{clean}}(k)$ is the power spectral magnitude of the reference signal at frequency bin k .

4. Results and Analysis

4.1. Baseline comparison

To evaluate the performance of our approach, we compared it with several noise-robust BWE methods. Table 1 presents the speech quality objective evaluation results on the Valentini-Botinhao noisy test set. The results indicate that our method achieves higher DNSMOS Overall and Background Noise scores compared to other state-of-the-art methods. While our

Table 1: *DNSMOS and LSD results of Noise-robust BWE models on Valentini-Botinhao noisy test set downsampled to 8KHZ*

Method	CSIG $\uparrow$	CBAK $\uparrow$	COVL $\uparrow$	LSD $\downarrow$
UEE [22]	2.27	2.39	2.17	2.72
MTL-MBE [23]	2.64	3.21	2.46	2.29
EP-WUN [24]	3.50	2.94	2.86	1.23
I-DTLN+ [25]	2.63	2.87	2.18	1.54
NU-Wav2 [9]	3.11	2.67	2.46	1.13
SDNet [26]	3.29	3.32	2.92	1.16
Proposed	3.28	4.08	3.04	1.54

Table 2: *Comparison results of different training strategy on Valentini-Botinhao noisy test set*

Method	CSIG $\uparrow$	CBAK $\uparrow$	COVL $\uparrow$	LSD $\downarrow$
Clean	3.51	4.04	3.21	-
Quantized	3.22	4.07	2.98	1.63
End2End	2.99	4.02	2.77	1.75
Progressive	3.17	4.06	2.94	1.61
Proposed	3.28	4.08	3.04	1.54

approach is generally competitive, it appears to aggressively suppress background noise, which can negatively impact the speech component in low signal-to-noise scenarios. This can explain why our method shows a notably higher background noise score alongside a lower signal score.

Figure 4 presents a comparison of spectrograms, demonstrating that the proposed method effectively restores the speech structure that is heavily masked by ambient noise. It outperforms the non-generative approaches, such as CSFNet[27].

4.2. Training strategy comparison

We investigated several training strategies using the same model architecture to examine the impact of different input features and training methods on performance. Table 2 compares three approaches:

- **Quantized:** In this approach, we use quantized embeddings as the input feature to predict clean tokens.
- **End2End:** Here, we forgo the two-stage training method (where the model is first pre-trained on a denoising task) and instead directly use downsampled noisy speech to predict high-resolution clean speech tokens.
- **Progressive:** In this method, we still use pre-quantized embeddings and a two-stage training strategy; however, the model is trained progressively, as described in Section 3.2.

According to Table 2, model performance significantly drops when the two-stage training method is omitted. This indicates that jointly training noise suppression and bandwidth expansion using a simple prediction model is more challenging and may struggle to converge to an optimal solution. In other words, It also implies pre-training on noise suppression provides a solid foundation for extending the model’s ability to perform bandwidth expansion. Because theoretically, bandwidth expansion rely on a relative clean speech component in low frequency range. Another key observation is that pre-quantized (or continuous) embeddings outperform quantized embeddings for the speech enhancement task in our work. This suggests that, al-

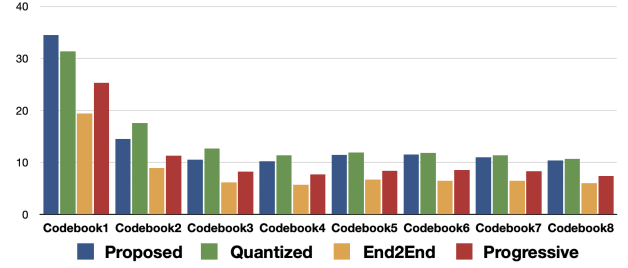


Figure 3: *Comparison of token prediction accuracy(%). Codebook1 represents the first codebook index prediction accuracy.*

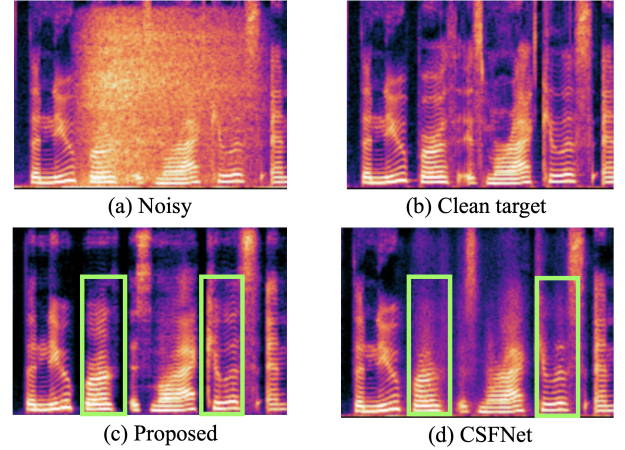


Figure 4: *Spectrograms comparison for restoring speech.*

though the neural codec performs well in signal reconstruction, some information is still lost during the compression process. As a result, using continuous embeddings as input features makes it easier for the model to predict clean tokens. Unexpectedly, the progressive training method did not outperform regular parallel prediction training. This suggests that our prediction model may benefit from predicting all tokens simultaneously. Figure 3 shows a bar graph of token prediction accuracy for each codebook using different training strategies. A joint analysis of Table 2 and Figure 3 reveals that the accuracy results are highly correlated with the objective evaluation metrics. In addition, quantized embeddings yield higher accuracy than our proposed method in all codebooks except codebook 1, confirming that codebook 1 plays a dominant role in speech reconstruction.

5. Conclusion

In this paper, we propose a neural codec approach for noise-robust bandwidth expansion that recovers high-resolution audio from low-resolution input under noisy conditions. Our experimental results demonstrate that our method outperforms other state-of-the-art techniques. Furthermore, our study on training strategies shows that a noise suppression pre-training process significantly benefits the bandwidth expansion task in noisy environments. Moreover, token prediction accuracy further confirms that the first codebook is the most critical component in neural codec-based speech generation. Future research will focus on incorporating semantic tokens to further enhance speech quality and mitigate speech suppression in low SNR scenarios.

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